Band-1 Hadamard Order 8564 – Miyamoto (Paley-II Seed H_2140 q=1069 + C_2142) + Paley-I Classical SPK; Gap 2052 Still OPEN

H_2140 Paley-II PASS; C_2142 PASS; H_8564 Miyamoto PASS; H_8564 Paley-I PASS -> IN SPK; Paley-II/Kronecker FAIL; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
Parents. `2026-09-16_Band1_Hadamard_8516_Miyamoto`; `2026-09-16_Band1_Hadamard_8356_PaleyII`; `2026-09-16_Band1_Hadamard_8452_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_8564_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 2140 and 8564 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Paley-II seed H_{2140} = Paley-II($q = 1069$) and symmetric Paley conference C_{2142} over GF(2141) ($q = 2141 \equiv 1 \pmod 4$), **and** classical Paley-I H_{8564} = Paley-I($q = 8563$). Classical SPK for 8564: Paley-I **PASS**es ($q = 8563$ prime $\equiv 3 \pmod 4$; verified); Paley-II **FAIL**s ($q = 4281 = 3 \cdot 1427$ NOT prime power – no matrix attempt); Kronecker **FAIL**s (4×2141). Therefore order **8564 is IN SPK via Paley-I** (Miyamoto is a parallel finite construction; Miyamoto alone never folds SPK). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under SPK. `clay_closed = false`. Prior ship: **8516 Miyamoto-only** (seeds **1064+2128** folded). Prior SPK seeds **1064+2128 / 2112 / 2088+8356 / 2028+8116 / 1996 / 1932 / 7604 / 1876+7508 / 1872 / 1860 / 7444 / 1848 / 7396 / 1788 / ...** retained; Miyamoto-only bonuses **8516/8452/7988/7732/7492/7156/7108/6836/6788/6676/6484/6404/6388/6196/...** excluded; **2140** folded into ONBOX/SPK when verified; **8564** folded via classical Paley-I (not Miyamoto alone).

Abstract

Finite constructions of orders **2140** and **8564**.

- H_{2140} = Paley-II ($q = 1069$) – preferred Miyamoto seed
- C_{2142} = Paley conference over GF(2141), $q = 2141$
- H_{8564} = Miyamoto(H_{2140}, C_{2142})
- H_{8564} = Paley-I ($q = 8563$) – classical SPK
- Paley-II for 8564: **SKIP** ($q = 4281$ NOT prime power; no matrix attempt)
- Kronecker for 8564: **SKIP** ($4 \cdot 2141$ not admissible)

SPK: 8564 IN SPK via Paley-I. Seed 2140 enters SPK (Paley-II). Prior SPK **1064+2128 / 2112 / 2088+8356 / ...** retained.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_8564.py`

Results. overall_pass=true; units 154/154; wall 1060.4434s; all max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 8564 = 4 \cdot 2141$; 2141 prime $\equiv 1 \pmod 4$ -> Miyamoto-eligible.
- Need H_{2140} and C_{2142} (Paley conference over prime field $q = 2141$).
- Seed: Paley-II $q = 1069$ (prime $\equiv 1 \pmod 4$) -> H_{2140} (Paley-I direct $q = 2139$ FAIL; Double FAIL; Kronecker 4×535 FAIL).
- SPK: Paley-I $q = 8563$ **PASS** (prime $\equiv 3 \pmod 4$; verified); Paley-II $q = 4281$ **FAIL** (NOT prime power); Kronecker **FAIL** -> **IN SPK via Paley-I**.
- Prior: 8516 Miyamoto-only shipped; this ship $q = 2141 \to n = 8564$.

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_2140 Paley-II	Paley-II q=1069 preferred seed	0	5.1869
C_2142	Paley conf GF(2141) q=2141	0	6.3722
H_8564 Miyamoto	Miyamoto(H_2140 Paley-II, C_2142)	0	498.1544
H_8564 Paley-I	Paley-I q=8563 classical SPK	0	538.9844
H_8564 Paley-II	SKIP q=4281=3*1427 (no matrix)	–	–
H_8564 Kronecker	SKIP 4*2141 not admissible	–	–

SPK closure note

Seeds include prior SPK ships through **8516** lineage (**1064+2128**; **2112**; **2088+8356**; **2028+8116**; **1996**; **1932**; **7604**; **1876/7508**; **1872**; **1860/7444**; **1848/7396**; **1788**; **...**; **NOT** 8516/8452/7988/7732/7492/7156/7108/6836/6788/6676/6484/6404/6388/6196) + verified **2140** (Paley-II) + verified **8564** (Paley-I classical); **exclude** Miyamoto-only bonuses. Next constructive gap **2052** (OPEN).

Quantity	Value
8564 in SPK	**True** (Paley-I q=8563)
2140 in SPK	**True** (Paley-II q=1069)
8516 in SPK	**False** (Miyamoto-only prior)
Next gap	**2052** (OPEN)
Miyamoto-only this ship	**False**

Next optional

q=2161->n=8644 Kronecker(4x540) OR q=2197->n=8788 PaleyII(q=1097) OR q=2209->n=8836 PaleyI(q=2207)+Double(PaleyI half=1104) OR method-attack SPK gap 2052 OR Track B / Object 1 meter await

Honesty gates

- `clay_closed = false`
- GATE-04 collection **LOCKED**
- Hadamard conjecture **OPEN**
- Gap 2052 **OPEN** (method triage / non-finding ONLY)
- No RH / P!=NP / YM / Frankl / NS / AC / completed-ToE claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)